



INBOUND SURVEILLANCE — Customer Quick-Start Guide

AI Bay Occupancy Monitoring · Technician Wrench-Time Analytics · Face ID · Telegram Dispatch

1. System Overview & Core Capabilities

Inbound Surveillance transforms your security cameras and webcams into proactive AI edge agents. Unlike passive CCTV that only records video, Inbound actively tracks vehicle bay occupancy, calculates technician wrench-time, records automated Face ID attendance, and sends instant Telegram photo proofs during safety breaches.

- **Local Edge Processing:** All video analysis runs 100% locally on your venue station. Video feeds never upload to external public clouds.
- **Two Interfaces Available:** Use the **Web Operations Console** in any browser (<http://localhost:5173/dashboard.html>) or launch the standalone **Native Desktop App**.

2. Navigating the Live View Console

The **Live View** is your day-to-day command center. It shows real-time camera video streams, active telemetry, and intelligent bay overlays:

- **Multi-Camera Grid:** Displays all active cameras. Click any tile to inspect it in full screen.
- **Top Telemetry Banner:** Displays *Ingest FPS* (camera delivery speed), *Infer Latency* (AI processing speed, typically 20-45ms), and resolution.
- **Audio Chimes (SOUND ON / MUTE):** Click the sound pill in the header to enable pleasant audio chimes whenever a car enters a service bay.
- **Shop Operating Hours:** Automatically marks the venue as **SHOP OPEN** during business hours (e.g. 08:00–18:00) and switches to **AFTER HOURS** mode (which triggers high-priority trespass alerts).

3. Understanding Service Bay Overlays & Colors

Border Color	Status Name	Detection Meaning	Operational Action
Vibrant Green	Active Labor	Vehicle present and technician actively detected working (wrench pose).	Accumulates productive wrench-time on technician scorecards.
Warm Amber	Bay Idle	Vehicle present, but no active tool movement detected for >120 seconds.	Increments idle timer; alerts manager to parts waiting or downtime.
Subtle Gray	Available / Empty	No vehicle or technician detected inside the service bay boundary.	Flags bay as open for immediate customer intake.

4. Setting Up Service Bays & ROIs (Drag-and-Drop)

To position service bays over your vehicle lifts or alignment racks:

1. Open **Live View** and locate your target camera feed.
2. Click the **Settings Cog** or right-click any existing bay to open the **Bay Context Menu**.
3. Select **Edit ROI** to activate interactive bounding handles. Drag the corners to frame your vehicle bay.
4. Assign a descriptive name (e.g. *Lift Bay 1*, *Tire Station*) and click **Save**.

5. Connecting Cameras & Video Sources

Inbound Surveillance supports real-time IP cameras, mobile phone cameras, USB webcams, and pre-recorded video files. Click **Add Camera** in the sidebar or setup modal to connect a video source.

Source Type	Protocol	Connection URL Example / Steps
IP Camera (RTSP) Hikvision, Dahua, Axis	rtsp	rtsp://username:password@192.168.1.50:554/h264Preview_01_main <i>Use camera's local IP address, RTSP port 554, and ONVIF/RTSP credentials.</i>
Pre-Recorded Video Uploaded MP4/AVI file	video	Select Local Video File , click Upload Video , and choose your file. <i>The engine loops the video locally for offline testing, demos, and forensics.</i>
Android Phone Using <i>IP Webcam</i> app	phone	http://192.168.1.85:8080/video <i>Ensure phone is on the same Wi-Fi. Start server in the app and enter the video URL.</i>
USB Webcam Logitech / PC Camera	webcam	Enter device index 0 (or 1 for secondary webcam, /dev/video0 on Linux). <i>Great for front-desk enrollment and test setups.</i>
One-Click Auto Scan ONVIF & Tapo Discovery	onvif / tapo	Click Scan Network . The system automatically detects compatible ONVIF and Tapo cameras on your subnet. Click Connect to apply.

How to Upload & Analyze Pre-Recorded Video Files

If you want to test the system or analyze pre-recorded incident footage without a live camera:

1. Click **Add Camera** in the sidebar or open **Settings** → **Source**.
2. Set **Protocol** to **Local Video File (MP4/AVI)**.
3. Click **Upload Video** and select your file (e.g., garage_shift_morning.mp4).
4. The system automatically uploads the file to the local edge directory (edge/videos/) and loops playback seamlessly through the AI pose and bay tracking pipeline.
5. Enter a camera label (e.g., *Shift Review - Bay 1*) and click **Connect & Save**.

Dual-Stream Best Practice (Pro Tip for Commercial Garages)

To get 25+ FPS AI detection without overloading your CPU, configure dual streams in config.yaml:

- **Substream (source:):** Set to 640x360 or 1280x720 resolution @ 15 FPS. Used for continuous, high-speed AI inference.
- **Mainstream (main_source:):** Set to 1080p or 4K resolution. Queried only when an alert fires to capture crisp photographic evidence.

PHONE STREAM SETUP

Phone Camera Tip: When using an Android phone with the free *IP Webcam* app, make sure to append /video to the end of the URL (e.g. http://192.168.1.85:8080/video). Avoid using the web interface root URL.

6. Connecting Your Telegram Alerts (1-Click Pairing)

Inbound Surveillance comes with an official pre-configured Telegram Bot. You do **not** need to create a bot or paste API tokens. The software automatically links your Telegram account and Chat ID directly to your signed-in profile with one click:

Step	Action	Exact Instructions
1	Sign In	Log in to your venue account on the Inbound Surveillance web console or desktop application so the bot knows which account to pair.
2	Click Connect	Navigate to Settings → Integrations (or open the Telegram Panel on the right sidebar) and click the green Connect Telegram button.
3	Tap Start	The official Inbound Surveillance bot opens automatically in your Telegram app. Tap Start (or send /start). Your unique pairing token is verified instantly.
4	Auto-Linked	Your Chat ID is automatically saved to your software account. The badge switches to Linked , activating instant incident photo dispatches and daily scorecards.
5	Send Test	Click Capture & Send Test (or Submit Ticket) in the panel to receive an immediate live photo proof on your smartphone and confirm delivery.

7. Quick Troubleshooting Matrix

Issue / Symptom	Probable Cause	Quick Fix (1 Minute)
Black screen or 'Connecting...'	Camera IP changed or wrong password.	Verify camera IP in router DHCP table. Test stream URL in VLC player.
Telegram alerts not sending	Telegram account not yet linked.	Open Settings → Integrations , click Connect Telegram , and tap Start in the Telegram app.
Frame rate drops below 10 FPS	Stream resolution too high for CPU.	Switch camera to 720p or 360p substream. In config.yaml set detect_fps: 15.
Technician marked 'Idle' under car	Worker occluded longer than grace limit.	Increase under_car_grace_seconds: 45 in config.yaml to maintain working state.
Face ID identifies staff as 'Customer'	Enrolled photo has poor lighting/angle.	Upload 2 fresh, front-facing photos in good lighting via Identity → Enroll Staff .

8. Desktop Application Downloads (Windows & Linux Ubuntu)

Operating System	Distribution Format	Download & Setup Instructions
Windows 10 / 11 64-bit Desktop Client [Experimental]	GitHub Actions Artifact (ZIP Package)	Download official Windows client installer build from GitHub: Click Here: Download Windows Build (GitHub Artifacts) URL: https://github.com/sothunly-alt/lbound-surveillance-V2/actions/runs/34489388811/artifacts/10157584518 Extract ZIP package and launch installer to complete setup.
Linux Ubuntu 20.04 / 22.04 / 24.04 [Verified & Stable]	Debian Package (.deb installer)	For Ubuntu Linux, the native .deb installer file is provided directly in the application folder. Install via terminal: <code>sudo dpkg -i Inbound_Surveillance_amd64.deb</code> (or double-click the .deb file to install via Ubuntu Software). Fully tested and error-free.

PRE-CONFIGURED LOGIN CREDENTIALS

Once installed, launch the application and log in using the pre-configured credentials:

• **Email:** testaccount@gmail.com | • **Password:** abcd1234

These credentials grant full access to live camera feeds, service bay overlays, and analytics.

9. Customer Support

INBOUND CREW SUPPORT

For system onboarding, camera configuration assistance, or technical inquiries, contact the Inbound Crew:

• **Support Email:** inboundcrew82@gmail.com

• **Service Hours:** Dedicated assistance and priority support for commercial garage operators.